CPE314 Data Communication and Computer Networks

Lecture 4: Digital Transmission

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Lecture Agenda

1. Introduction to Digital Signals
2. Baseband Transmission
3. Broadband Transmission
4. Summary

Digital Signals

Two-Level vs Multi-Level Signals

A digital signal can have:

- **Two levels** (Binary: 0 and 1)
- **Multi-level** (More than two voltage levels)

- If a signal has L levels:

$$L = 2^n$$

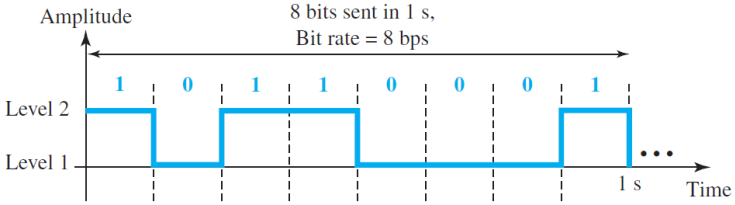
- Number of bits per level:

$$n = \log_2 L$$

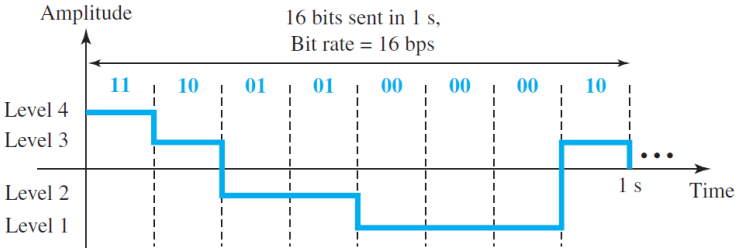
Important

Increasing levels increases bits per signal element.

Digital Signal Levels



a. A digital signal with two levels



b. A digital signal with four levels

Bit Rate

Definition

Bit rate is the number of bits transmitted per second (bps).

- Digital signals are usually nonperiodic
- Period and frequency are not suitable descriptors
- Bit rate replaces frequency for digital signals

Key Insight

Bit rate determines bandwidth requirement.

Bit Rate Calculation Examples

Text Download

- 100 pages/sec
- 24 lines/page
- 80 char/line
- 8 bits/char

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$$100 \times 24 \times 80 \times 8$$

$$= 1.536 \text{ Mbps}$$

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Digitized Voice

- Bandwidth $= 4$ kHz
- Sampling $= 2 \times 4000$
- 8 bits/sample

Bit Rate Calculation Examples

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Digitized Voice

- Bandwidth = 4 kHz
- Sampling = 2×4000
- 8 bits/sample

$$2 \times 4000 \times 8$$

$$= 64 \text{ kbps}$$

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Digitized Voice

- Bandwidth $= 4$ kHz
- Sampling $= 2 \times 4000$
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HDTV

- 1920×1080
- 30 frames/sec
- 24 bits/pixel

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Digitized Voice

- Bandwidth $= 4$ kHz
- Sampling $= 2 \times 4000$
- 8 bits/sample

$$2 \times 4000 \times 8$$

$$= 64 \text{ kbps}$$

HDTV

- 1920×1080
- 30 frames/sec
- 24 bits/pixel

$$1920 \times 1080 \times 30 \times 24$$

$$\approx 1.49 \text{ Gbps}$$

Observation

- Different applications require different bit rates — from kbps (voice) to Gbps (video).
- Image/Video compression can be used if required.

Bit Length

Definition

Bit length = distance one bit occupies on the medium

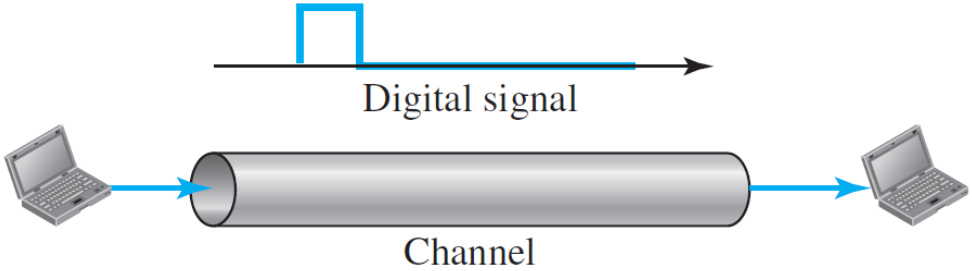
$$\text{Bit Length} = \text{Propagation Speed} \times \text{Bit Duration}$$

Baseband Transmission

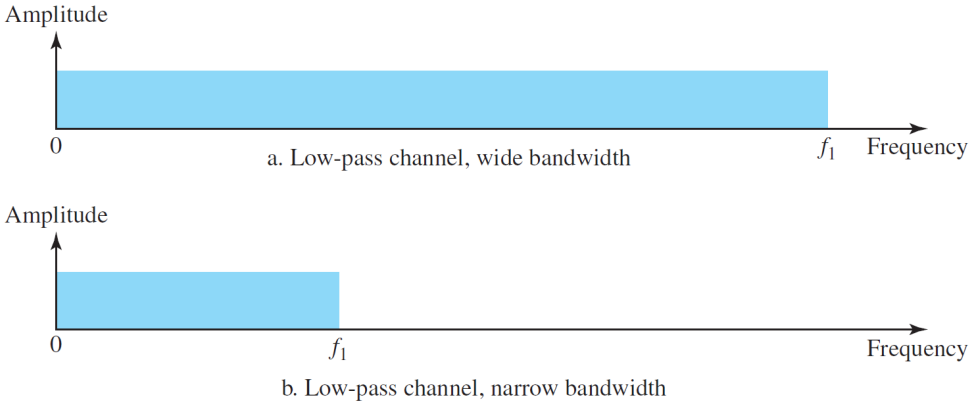
Definition

Sending digital signal directly over a low-pass channel.

- Channel bandwidth starts from 0 Hz
- No modulation required
- Used in LAN cables



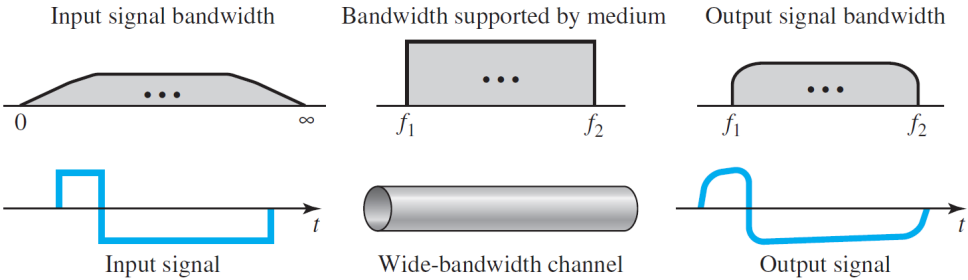
Low-Pass Channel: Wideband vs. Narrowband



a. Low-pass channel, wide bandwidth

b. Low-pass channel, narrow bandwidth

Wide Bandwidth Case



Limited/Narrow Bandwidth – Rough Approximation

Worst Case Pattern:

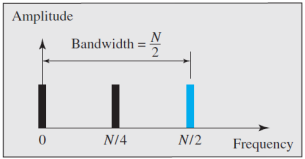
01010101...

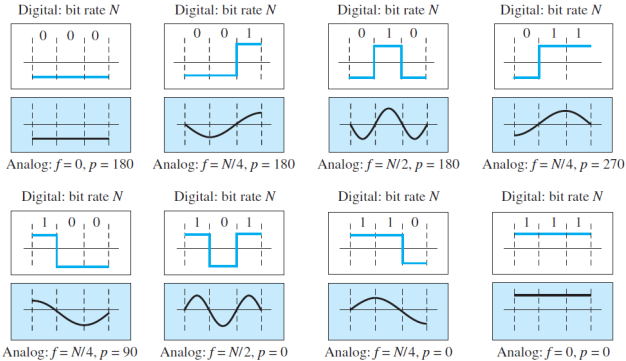
- Maximum transitions
- Highest required frequency

$$f = \frac{N}{2} \implies B = \frac{N}{2} - 0 = \frac{N}{2}$$

Key Idea

Using only the first harmonic.





Limited/Narrow Bandwidth – Better Approximation

Adding Higher Harmonics

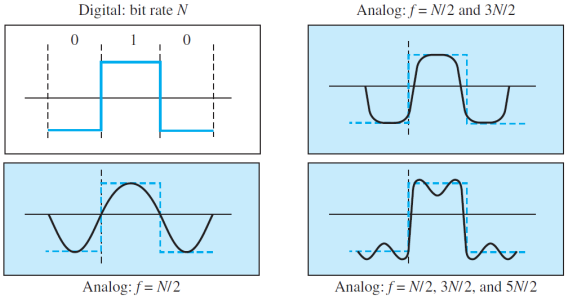
- $B = \frac{3N}{2}, \frac{5N}{2}, \frac{7N}{2}, \ldots$
- Even harmonics cancel
- Usually 5th harmonic is considered the highest

Observation

- Bandwidth increases with number of harmonics.

Important

Required bandwidth $\propto$ Bit rate



Practice Problems: Baseband Transmission

Problem 1

What is the required bandwidth of a low-pass channel if we need to send **1 Mbps** using baseband transmission?

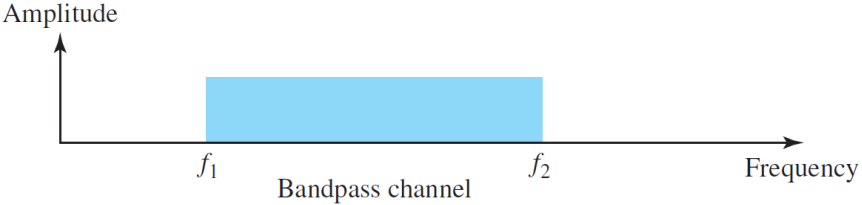
Problem 2

We have a low-pass channel with bandwidth **100 kHz**. What is the maximum bit rate of this channel?

Bandpass Channel

Definition

Bandwidth does not start at 0 Hz.

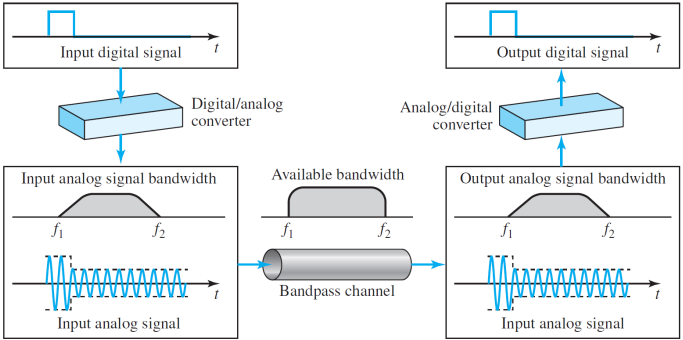


Modulation

Definition

Digital signal converted to analog before transmission.

- Uses carrier signal
- Amplitude / Frequency / Phase modified



Example: Telephone Line

- Bandwidth: 0–4 kHz
- Baseband max $\approx$ 8 kbps
- Solution: Use modem

Modem

Modulator + Demodulator

Example: Digital Cellular System

- Voice digitized
- Channel divided among many users
- Requires modulation

Important

If channel is bandpass, modulation is mandatory.

Summary

- Digital signals can be two-level or multi-level
- Bit rate replaces frequency for digital signals
- Baseband requires low-pass channel
- Required bandwidth $\propto$ bit rate
- Bandpass channels require modulation

Reference

B. A. Forouzan, *Data Communications and Networking*, 5th ed. New York, NY, USA: McGraw-Hill, 2013.